TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Specific Heat of a Liquid by the Method of Cooling

No heater, no hot solid — just a stopwatch. The liquid that cools fastest through the same temperature interval is the one holding the least heat.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 14 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the specific heat s_l of a liquid (here kerosene) by comparing its cooling time with water's in the same calorimeter. By **Newton's law of cooling**, a body whose surface, surroundings and temperature excess are fixed loses heat at a definite rate. Fill the calorimeter with hot water and time its fall through a chosen interval, say 60 °C → 55 °C: the heat surrendered is $(m_w + W)\Delta\theta$ in time t_w. Repeat with the same mass of the liquid in the same calorimeter, through the same interval, in the same surroundings: the heat surrendered is $(m_l\, s_l + W)\Delta\theta$ in time t_l. The calorimeter's surface and the temperature range are unchanged, so the average rate of loss is the same in both runs:

$$(m_w + W) / t_w = (m_l\, s_l + W) / t_l \quad \Rightarrow \quad s_l = [\, (m_w + W)(t_l/t_w) - W \,] / m_l$$

The method never measures a quantity of heat at all — it measures two times. That is its elegance and its demand: everything except the liquid must be held identical between the runs, because the equality of loss rates is the entire argument.

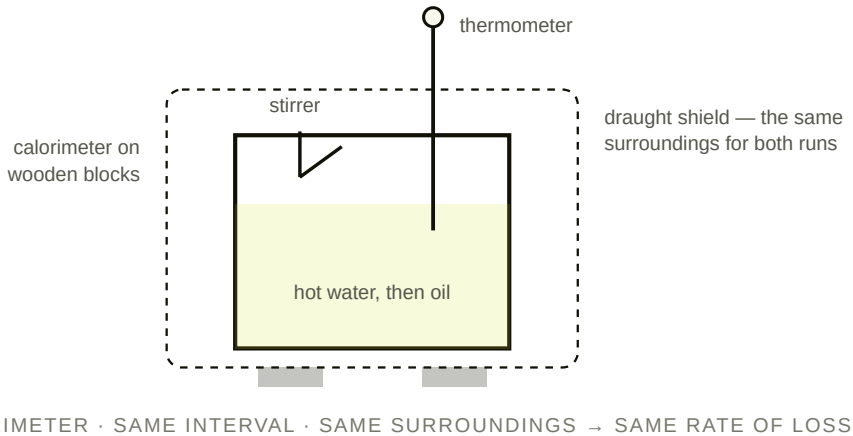


Figure 1 — The cooling arrangement. Nothing is heated during a run: the liquid simply cools, shielded from draughts, while the stopwatch times its passage through the chosen temperature interval.

2. Apparatus and procedure

- **Apparatus:** calorimeter with stirrer and lid inside a draught shield, thermometer (0.1 °C), stopwatch, balance, hot water, the liquid under test, a kettle for heating.
- Weigh the calorimeter (for W). Add a measured mass of hot water (heated to about 70 °C), m_w. Stir, and start the stopwatch as the temperature falls through 60 °C; stop it at 55 °C. Record t_w. (Better: record the whole cooling curve — temperature every minute — and read both times off the graph, as in Figure 2.)
- Empty, dry and refill the calorimeter with the *same mass* of the liquid, $m_l = m_w$, heated to the same starting temperature. Time the same 60 °C → 55 °C fall: t_l.
- Keep everything else untouched — same shield, same room conditions, same exposed surface, lid on, gentle stirring in both runs.
- Compute s_l from the ratio of the times.

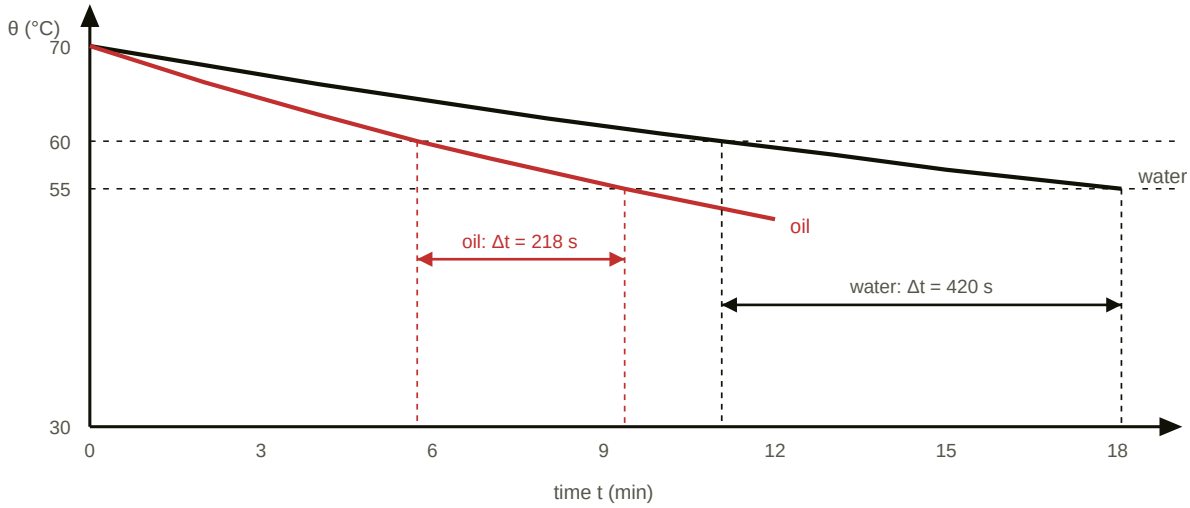


Figure 2 — Cooling curves for the two runs of the worked example (room temperature 30 °C). Both curves cross the same 60 °C → 55 °C band; the oil crosses it in 218 s against water's 420 s — its smaller heat content drains away at the same rate, so it spends less time in the band.

3. Worked calculation

Calorimeter water equivalent W = 10 g. Mass of each liquid: 150 g. Timed interval 60 °C → 55 °C: water takes t_w = 420 s; kerosene takes t_l = 218 s.

$$s_l = [\ (150 + 10)(218/420) - 10\ ] / 150 = [\ 83.0 - 10\ ] / 150 = 73.0 / 150$$

$$s_l = 0.49 \text{ cal g}^{-1} \text{°C}^{-1} \approx 2.0 \times 10^3 \text{ J kg}^{-1} \text{K}^{-1}$$

The accepted value for kerosene is about 0.50 cal g^{-1} $°C^{-1}$. The check on the reasoning: 160 × 218/420 = 83.0 — the oil-plus-calorimeter behaves, thermally, like 83 g of water; strip out the calorimeter's 10 g and 73 g of "water-equivalent heat capacity" remains for 150 g of oil, hence about half of water's specific heat, gram for gram.

4. Precautions

- Identical conditions are the experiment: same calorimeter, same mass, same interval, same shield, same room. A draught during one run only is a second, invisible variable.
- Use the same temperature interval for both runs — Newton's law makes the loss rate depend on the temperature excess, so equal intervals are what make the rates equal.
- Stir gently and equally in both runs; stirring changes the surface renewal and hence the loss rate.
- Start both runs from the same temperature and use freshly heated liquid each time; a volatile liquid also loses mass between runs, and m_l must be the mass actually present.

Practice (viva preparation)

1. State Newton's law of cooling and the two conditions this experiment borrows from it.
2. In the same calorimeter, 150 g of water cools from 60 °C to 55 °C in 300 s. The same volume of a liquid (mass 120 g, density 0.8 g cm^{-3}) takes 150 s over the same interval. W = 10 g. Find the liquid's specific heat.
3. Why is the same *mass* commonly used, yet the answer key above works with the same *volume* — what changes in the formula?
4. A student times the water from 70 °C to 65 °C but the oil from 60 °C to 55 °C. Why is the comparison worthless?
5. Why does this method need no thermometer calibration beyond reading two fixed temperatures consistently — and no knowledge of the heat actually lost?

Answer key

1. Newton's law: the rate of heat loss of a body is proportional to its excess temperature over the surroundings (for small excesses and unchanged surface). The experiment borrows (i) that the rate depends only on the excess temperature and the surface/surroundings, and (ii) that over the same interval, in the same apparatus, the average rate is therefore the same for both liquids.
2. $(m_w + W)(t_l/t_w) = 160 \times 150/300 = 80$. So $m_l\, s_l = 80 - 10 = 70$, and $s_l = 70/120 = 0.58$ cal g^{-1} $°C^{-1}$.
3. Nothing structural: the formula needs each liquid's actual mass. Equal masses make $m_l = m_w$ and the arithmetic shorter; equal volumes (same filling level, same exposed surface) are arguably better physics, but then the masses differ by the density ratio and must be entered separately.
4. The loss rate grows with the temperature excess. The 70 → 65 °C run cools against a larger excess, so it loses heat faster per degree than the 60 → 55 °C run; comparing its time with the other's mixes two different rates, and the equality the method stands on collapses.
5. Because only a ratio of times enters the final formula. The heat lost per second cancels between the two runs, and the thermometer is used only as a marker for the same two temperatures in both — a constant thermometer error shifts both runs together and cancels.